

Worked Example

Joke Generator

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Step 1 – The Idea

- We want a small program that can:
 1. Tell a random joke (no input needed).
 2. Tell a funny story using a **name** provided by the user.
 3. Take in an **age**, calculate the age in 10 years, and return a jokey message.



Step 2 - Designing the Methods

Method Name	Parameters	Return Type	Description
<code>tellJoke()</code>	<code>none</code>	<code>void</code>	Prints a random joke.
<code>tellStory(String name)</code>	<code>name</code>	<code>void</code>	Uses the name to tell a funny story.
<code>ageInTenYears(int age)</code>	<code>age</code>	<code>String</code>	Returns a message about the person's future age.

Step 3 - Pseudocode

- **JokeGenerator** class:
 - **tellJoke:**
Print a simple programmer joke.
 - **tellStory(name):**
Print a short funny story that uses the person's name.
 - **ageInTenYears(age):**
Work out age + 10.
Make a jokey message using that number.
Give the message back to whoever called it.
- **Driver** class:
 - Start the program.
 - Make a new JokeGenerator object called joker.
 - Call joker's tellJoke method.
 - Ask the user for their name.
 - Call joker's tellStory, passing in the name.
 - Ask the user for their age.
 - Call joker's ageInTenYears, store the message.
 - Print the message.
 - Program ends.

Step 4 - Java Code

```
public class JokeGenerator {

    // Method 1: No parameters, no return
    public void tellJoke() {
        System.out.println("Why don't programmers like nature? It has too many bugs!");
    }

    // Method 2: Parameter, no return
    public void tellStory(String name) {
        System.out.println("Once upon a time, " + name + " tried to debug code at 3am...");
        System.out.println("The bug fixed itself. Legends still talk about it.");
    }

    // Method 3: Parameter and return value
    public String ageInTenYears(int age) {
        int futureAge = age + 10;
        return "In 10 years, you'll be " + futureAge + " years old — still debugging!";
    }
}
```

```
import java.util.Scanner;
public class Driver {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Create an object of JokeGenerator
        JokeGenerator joker = new JokeGenerator();

        // 1. Tell a joke
        joker.tellJoke();

        // 2. Ask for name and tell a story
        System.out.print("\nEnter your name: ");
        String userName = input.nextLine();
        joker.tellStory(userName);

        // 3. Ask for age and give future message
        System.out.print("\nEnter your age: ");
        int age = input.nextInt();
        String message = joker.ageInTenYears(age);
        System.out.println(message);
    }
}
```

Lets Get Coding..

